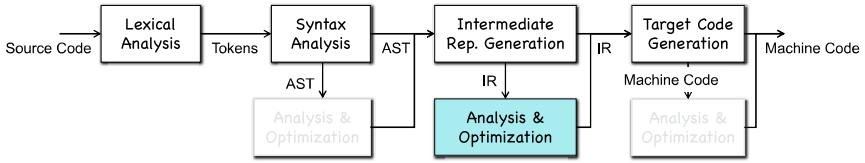
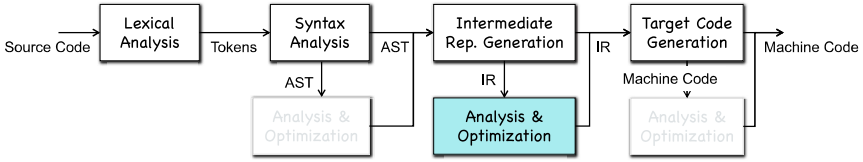
Chapter 9-2
Symbolic Execution

Data Flow Analysis



- Data flow analysis is a form of static program analysis
  - Computes dataflow facts held at a program point
  - E.g., what definitions can reach the 2nd basic block?
  - **Follow the order of control flows (order of statements/blocks)**

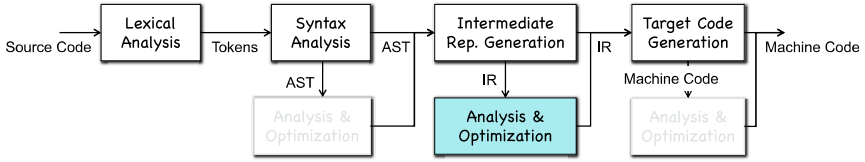
Data Flow Analysis



- Data flow analysis is a form of static program analysis
  - Computes dataflow facts held at a program point
  - E.g., what definitions **Flow Sensitive!** block?
  - Follow the order of c ments/blocks)

Is a data flow fact held at a program point?

Data Flow Analysis



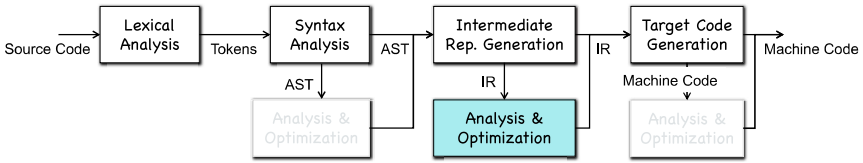
Is a data flow fact held ~~at a program point~~?

Flow Insensitive → **Flow Sensitive!** → **Path Sensitive**

Advantage: High Precision
Disadvantage: Path Explosion

Is a data flow fact held along a program path?

Data Flow Analysis



PART I: Symbolic Execution

Symbolic Execution

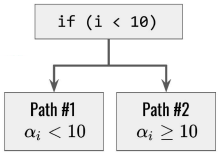
- Symbolic execution is a classic path-sensitive analysis that enumerates and analyzes each path of a program
  - Bug finding;
  - Software testing;
  - …

https://github.com/klee/klee | https://clang-analyzer.llvm.org/

```
clang -analyze -Xanalyzer -analyzer-checker=core.DivideZero hello.c
```


Symbolic Execution

- Each input variable is associated to a symbol, *e.g.*, `int i → α_i`
- Each program statement generate formulas over the input symbols, *e.g.*, `i * 2 + 5 → 2 * α_i + 5`
- Program path => path constraint



Symbolic Execution

```
1.  void foobar(int a, int b) {
2.     int x = 1, y = 0;
3.     if (a != 0) {
4.        y = 3+x;
5.        if (b == 0)
6.           x = 2*(a+b);
7.     }
8.     assert(x-y != 0);
9.  }
```

$\sigma = \{ a \rightarrow \alpha_a ; b \rightarrow \alpha_b ; x \rightarrow 1 ; y \rightarrow 0 \}$;
$\pi = true$

$\sigma = \{ a \rightarrow \alpha_a ; b \rightarrow \alpha_b ; x \rightarrow 1 ; y \rightarrow 0 \}$;
$\pi = \alpha_a = 0$
check: $\pi \wedge \neg(1 - 0 \neq 0)$ <=> false **ok**

σ: symbolic store that maps each variable to symbolic expressions.
π: path constraints. Initially, $\pi = true$.

Symbolic Execution

```
1.  void foobar(int a, int b) {
2.     int x = 1, y = 0;
3.     if (a != 0) {
4.        y = 3+x;
5.        if (b == 0)
6.           x = 2*(a+b);
7.     }
8.     assert(x-y != 0);
9.  }
```

$\sigma = \{ a \rightarrow \alpha_a ; b \rightarrow \alpha_b ; x \rightarrow 1 ; y \rightarrow 0 \}$;
$\pi = true$

$\sigma = \{ a \rightarrow \alpha_a ; b \rightarrow \alpha_b ; x \rightarrow 1 ; y \rightarrow 4 \}$;
$\pi = \alpha_a \neq 0$

$\sigma = \{ a \rightarrow \alpha_a ; b \rightarrow \alpha_b ; x \rightarrow 2(\alpha_a + \alpha_b) ; y \rightarrow 4 \}$;
$\pi = \alpha_a \neq 0 \wedge \alpha_b = 0$

check: $\pi \wedge \neg(2(\alpha_a + \alpha_b) - 4 \neq 0)$
satisfiable => error

σ: symbolic store that maps each variable to symbolic expressions.
π: path constraints. Initially, $\pi = true$.

Solving Path Constraints

- How can we check the satisfiability of a constraint?

$$\alpha_a \neq 0 \wedge \alpha_b = 0 \wedge \neg(2(\alpha_a + \alpha_b) - 4 \neq 0)$$

- Constraint solver
  - Satisfiable => A possible solution
  - Unsatisfiable
  - Unknown (due to timeout)

Z3 https://github.com/Z3Prover/z3

CVC5 https://cvc5.github.io

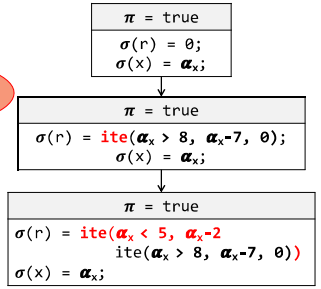
Limitations

- **Scalability**
  - (1) Path Explosion
  - (2) External Calls
  - (3) Constraint Solving
- **Optimizations**
  - (1) Path/State Merging
  - (2) Induction for Loops
  - (3) Concolic Execution
  - (4) Speculative Execution
  - (5) ……

Path (or State) Merging

```
1  void foo(int x) {
2      int r = 0;
3
4      if (x > 8)
5          r = x - 7;
6
7      if (x < 5)
8          r = x - 2;
9
10     assert(r != 5);
11 }
```

Avoid repetitive analysis!

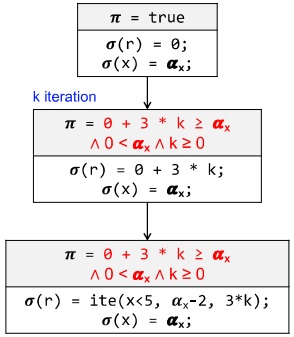


More complicated constraints!

Check if $\pi \wedge \sigma(r) = 5$ is satisfiable.

Induction for Loops

```
1  void foo(int x) {
2      int r = 0;
3
4      while (r < x)
5          r += 3;
6
7      if (x < 5)
8          r = x - 2;
9
10     assert(r != 5);
11 }
```



Check if $\pi \wedge \sigma(r) = 5$ is satisfiable.

Induction for Loops

```
1  void foo(int x) {
2      int r = 0;
3
4      while (r < x)
5          if (*) r += 3; else r *= 4;
6
7      if (x < 5)
8          r = x - 2;
9
10     assert(r != 5);
11 }
```

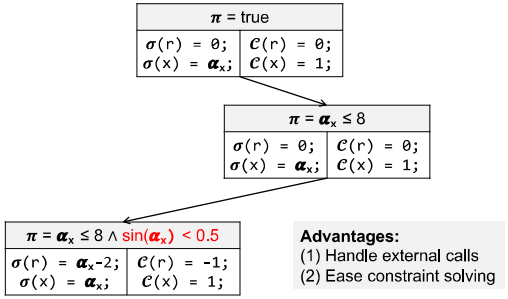
It does not always work!

Concolic Execution

- Concrete + Symbolic Execution
- Perform symbolic execution at runtime

Concolic Execution

```
1  void foo(int x) {
2      int r = 0;
3
4      if (x > 8)
5          r = x - 7;
6
7      if (sin(x) < 0.5)
8          r = x - 2;
9
10     assert(r != 5);
11 }
```



Advantages:
(1) Handle external calls
(2) Ease constraint solving

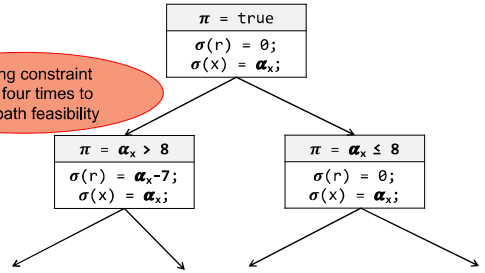
Check if $\alpha_x \leq 8 \wedge \sin(\alpha_x) < 0.5 \wedge \alpha_x - 2 = 5$ is satisfiable.

Replace sin(x) with the concrete value sin(1), i.e., 0.1

Speculative Execution

```
1  void foo(int x) {
2      int r = 0;
3
4      if (x > 8)
5          r = x - 7;
6
7      if (x < 5)
8          r = x - 2;
9
10     assert(r != 5);
11 }
```

Invoking constraint solver four times to check path feasibility



Speculative Symbolic Execution: Provide a way to guess path feasibility w/o constraint solving, thereby reducing the invocation of the constraint solver!

PART II: Satisfiability

Logic

- A logic is just a formal language (just like regular languages, etc.)
- (1) **Syntax**: What are the legal sentences of the language?
- (2) **Semantics**: What does a sentence mean?

Propositional Logic (PL)

- What are legal sentences of PL?
  - $\bot$
  - $\top \wedge \bot$
  - $p \vee q$
  - p **and** q
- How would you encode "If it is cold and rainy, then it is either warm or rainy" as a formula in PL?

Solving Path Constraints

- Propositional Logic
- Satisfiability (SAT) --- DPLL and CDCL

- First Order Logic
- Satisfiability Modulo Theories (SMT)
- The Bit Vector Theory

Propositional Logic (PL)

- **Syntax of PL**, i.e., what are legal sentences of PL?
- (1) **Truth Symbols (b)**: $\top$ and $\bot$ (true and false)
- (2) **Propositional Variables (v)**: $p,\ q,\ r,\ \ldots$

} Atoms

- (3) **Literals (l)**: atom or its negation
- (4) **Formulas (F)**: $l \mid \neg F \mid F_1 \vee F_2 \mid F_1 \wedge F_2 \mid F_1 \Rightarrow F_2 \mid F_1 \Leftrightarrow F_2$

Satisfiability and Validity

- A **solution** of a formula F in PL is a map I that maps each variable to either $\top$ or $\bot$
- A formula F in PL is **satisfiable**, iff there exists I: $I \vDash F$
- A formula F in PL is **valid**, iff for all I: $I \vDash F$

Theorem [Duality of Satisfiability and Validity]
For all formulae F in propositional logic, F is valid if and only if $\neg F$ is not satisfiable.

Normal Forms

- A normal form is a syntactic restriction on the shape of a formula
- A normal form structure the search space and simplify the decision procedure

Negation Normal Form (NNF)

- **Negation normal form** disallows implications and only allows variables to be negated

- Which of these formulas are in NNF?
  - $(p \wedge q) \Rightarrow (p \vee \neg q)$
  - $\neg\, p \vee \neg q \vee p \vee \neg q$
  - $\neg(p \wedge q) \vee p \vee \neg q$

Theorem [Sufficiency of NNF]
For all formulae F in propositional logic, there exists an equivalent formula in negation normal form.

Conjunctive Normal Form (CNF)

- Formulas in CNF are conjunctions of disjunctive literals

$$\wedge_i \vee_j l_{ij}$$

- Which of these formulas are in CNF?
  - $(p \wedge q) \Rightarrow (p \vee \neg q)$
  - $(\neg\, p \vee \neg q) \wedge (p \vee \neg q)$
  - $\neg p \vee \neg q \vee p \vee \neg q$

Theorem [Sufficiency of CNF]
For all formulae F in PL, there exists an equivalent formula in CNF.

Negation Normal Form (NNF)

- **Negation normal form** disallows implications and only allows variables to be negated
  - (1) **Truth Symbols (b)**: $\top$ and $\bot$ (true and false)
  - (2) **Propositional Variables (v)**: p, q, r, …
  - (3) **Literals (l)**: atom or its negation
  - (4) **Formulas (F)**: $l \mid \neg F \mid F_1 \vee F_2 \mid F_1 \wedge F_2 \mid F_1 \Rightarrow F_2 \mid F_1 \Leftrightarrow F_2$

Disjunctive Normal Form (DNF)

- Formulas in DNF are disjunctions of conjunctive literals

$$\vee_i \wedge_j l_{ij}$$

- Which of these formulas are in DNF?
  - $(p \wedge q) \Rightarrow (p \vee \neg q)$
  - $\neg\, p \wedge \neg q \vee (p \wedge \neg q)$
  - $\neg(p \wedge q) \vee p \vee \neg q$
  - $\neg p \vee \neg q \vee p \vee \neg q$

Size Explosion!

Theorem [Sufficiency of DNF]
For all formulae F in PL, there exists an equivalent formula in DNF.

Equisatisfiability

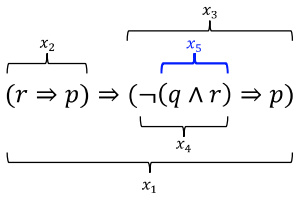
- Two formulas F_1 and F_2 are equisatisfiable iff:

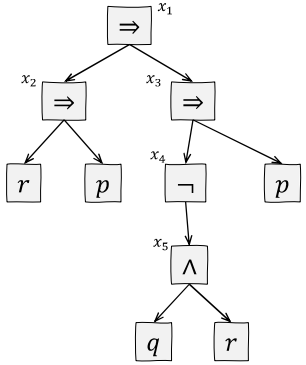
  F_1 is satisfiable $\Leftrightarrow$ F_2 is satisfiable

- Equisatisfiability ≠ Equivalence

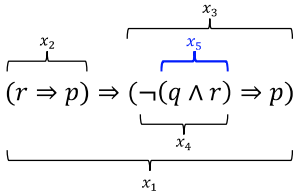
Theorem [Efficient CNF Encoding]
For all formulae F in PL, there exists an equisatisfiable formula in CNF that is at most a constant factor larger than F.

Tseitin Transformation





Tseitin Transformation



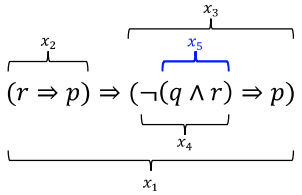
$$
\begin{aligned}
& x_1 \wedge (x_1 \Leftrightarrow (x_2 \Rightarrow x_3)) \\
\wedge\ & x_2 \Leftrightarrow (r \Rightarrow p) \\
\wedge\ & x_3 \Leftrightarrow (x_4 \Rightarrow p) \\
\wedge\ & x_4 \Leftrightarrow \neg x_5 \\
\wedge\ & x_5 \Leftrightarrow q \wedge r
\end{aligned}
$$

Theorem [Efficient CNF Encoding]
For all formulae F in PL, there exists an equisatisfiable formula in CNF that is at most a constant factor larger than F.

DPLL

- $(p \vee q \vee \neg r) \wedge (p \vee q \vee r) \wedge (p \vee \neg q) \wedge \neg p$
- **Goal**: Find a solution to satisfy the input formula
- Repeat the following steps:
- (1) Choose a variable, p, assign true or false to the variable
- (2) Propagate the value of p

Tseitin Transformation



$$
\begin{aligned}
& x_1 \wedge (x_1 \Leftrightarrow (x_2 \Rightarrow x_3)) \\
\wedge\ & x_2 \Leftrightarrow (r \Rightarrow p) \\
\wedge\ & x_3 \Leftrightarrow (x_4 \Rightarrow p) \\
\wedge\ & x_4 \Leftrightarrow \neg x_5 \\
\wedge\ & x_5 \Leftrightarrow q \wedge r
\end{aligned}
$$

$$x_4 \Rightarrow \neg x_5 \wedge \neg x_5 \Rightarrow x_4$$

$$(\neg x_4 \vee \neg x_5) \wedge (x_5 \vee x_4)$$

The SAT Problem

- Deciding satisfiability of a formula F in PL
- Find a solution such that we can rewrite F to be true

Recap: A **solution** of a formula F in PL is a map I that maps each variable to either $\top$ or $\bot$

DPLL

- $(p \vee q \vee \neg r) \wedge (p \vee q \vee r) \wedge (p \vee \neg q) \wedge \neg p$

```
bool DPLL (F, I) {
    if (F == false) return UNSAT;
    if (F == true)  return SAT;

    p = choose(F);
    bool ret = DPLL(F[p→true], I[p→true])
    if (ret == SAT) return SAT;

    return DPLL(F[p→false], I[p→ false]);
}
```

DPLL

- $(p \vee q \vee \neg r) \wedge (p \vee q \vee r) \wedge (p \vee \neg q) \wedge \neg p$
- $(p \vee q) \wedge (p \vee \neg q) \wedge \neg p$
- $p \wedge \neg p$ **False! Unsatisfiable!**

Theorem [Resolution]

$(a_0 \vee a_1 \vee \cdots \vee a_n \vee c) \wedge (b_0 \vee b_1 \vee \cdots \vee b_m \vee \neg c)$

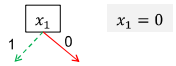
can be simplified into

$(a_0 \vee a_1 \vee \cdots \vee a_n \vee b_0 \vee b_1 \vee \cdots \vee b_m)$

CDCL

- Conflict-Driven Clause Learning
- Make DPLL faster by learning from mistakes

CDCL



$x_1 = 0$

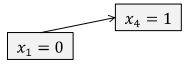
- $(x_1 \vee x_4) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
- $(x_2 \vee x_{11}) \wedge$
- $(\neg x_7 \vee \neg x_3 \vee x_9) \wedge$
- $(\neg x_7 \vee x_8 \vee \neg x_9) \wedge$
- $(x_7 \vee x_8 \vee \neg x_{10}) \wedge$
- $(x_7 \vee x_{10} \vee \neg x_{12})$

CDCL

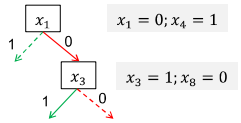
$x_1 = 0; x_4 = 1$

- $(x_1 \vee x_4) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
- $(x_2 \vee x_{11}) \wedge$
- $(\neg x_7 \vee \neg x_3 \vee x_9) \wedge$
- $(\neg x_7 \vee x_8 \vee \neg x_9) \wedge$
- $(x_7 \vee x_8 \vee \neg x_{10}) \wedge$
- $(x_7 \vee x_{10} \vee \neg x_{12})$

Implication Graph



CDCL



$x_1 = 0; x_4 = 1$

$x_3 = 1; x_8 = 0$

- $(x_1 \vee x_4) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
- $(x_2 \vee x_{11}) \wedge$
- $(\neg x_7 \vee \neg x_3 \vee x_9) \wedge$
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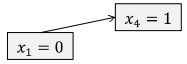

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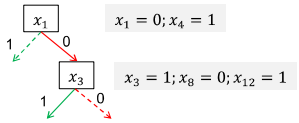

$x_1 = 0; x_4 = 1$

$x_3 = 1; x_8 = 0; x_{12} = 1$

- $(x_1 \vee x_4) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
- $(x_2 \vee x_{11}) \wedge$
- $(\neg x_7 \vee \neg x_3 \vee x_9) \wedge$
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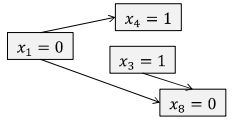
Implication Graph



CDCL

- $(x_1 \vee x_4) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
- $(x_2 \vee x_{11}) \wedge$
- $(\neg x_7 \vee \neg x_3 \vee x_9) \wedge$
- $(\neg x_7 \vee x_8 \vee \neg x_9) \wedge$
- $(x_7 \vee x_8 \vee \neg x_{10}) \wedge$
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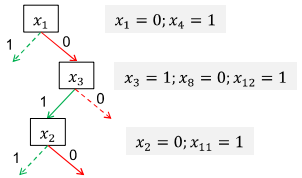

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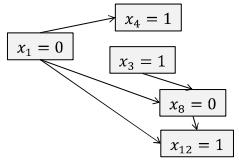
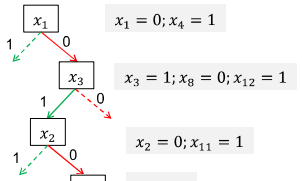
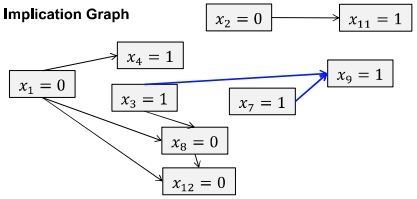


CDCL

- $(x_1 \vee x_4) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
- $(x_2 \vee x_{11}) \wedge$
- $(\neg x_7 \vee \neg x_3 \vee x_9) \wedge$
- $(\neg x_7 \vee x_8 \vee \neg x_9) \wedge$
- $(x_7 \vee x_8 \vee \neg x_{10}) \wedge$
- $(x_7 \vee x_{10} \vee \neg x_{12})$



Implication Graph



CDCL

- $(x_1 \vee x_4) \wedge$
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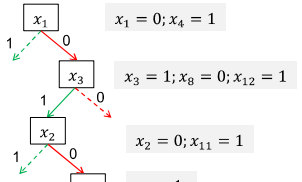
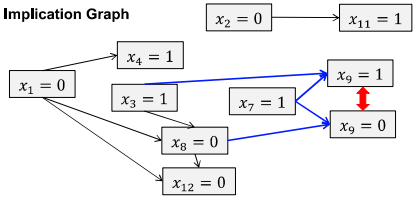
- $(x_7 \vee x_{10} \vee \neg x_{12})$



Implication Graph



CDCL

- $(x_1 \vee x_4) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
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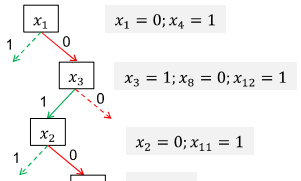
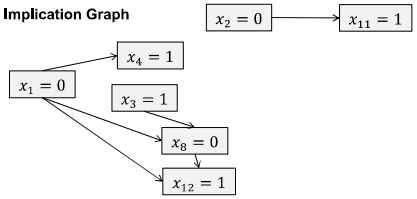


Implication Graph



CDCL

- $(x_1 \vee x_4) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
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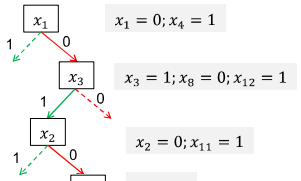
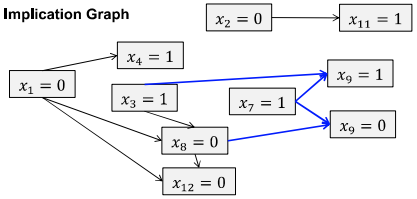


Implication Graph



CDCL

- $(x_1 \vee x_4) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
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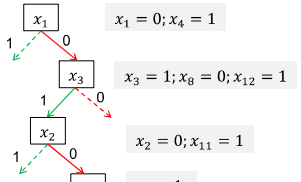
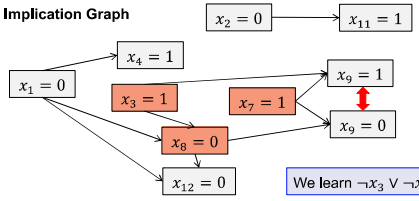
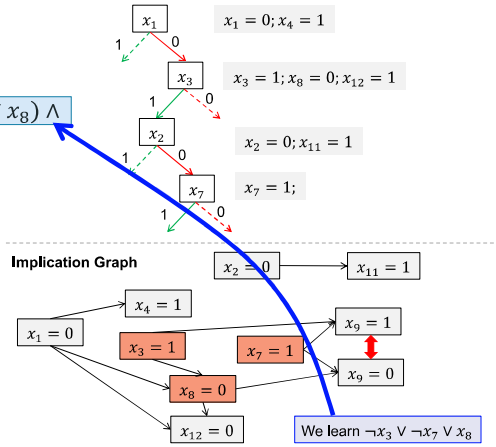


Implication Graph



CDCL

- $(x_1 \vee x_4) \wedge (\neg x_3 \vee \neg x_7 \vee x_8) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
- $(x_2 \vee x_{11}) \wedge$
- $(\neg x_7 \vee \neg x_3 \vee x_9) \wedge$
- $(\neg x_7 \vee x_8 \vee \neg x_9) \wedge$
- $(x_7 \vee x_8 \vee \neg x_{10}) \wedge$
- $(x_7 \vee x_{10} \vee \neg x_{12})$



Recap

Given a formula F in propositional logic, DPLL/CDCL can be used to decide its satisfiability!

CDCL

- $(x_1 \vee x_4) \wedge (\neg x_3 \vee \neg x_7 \vee x_8) \wedge$
- $(x_1 \vee \neg x_3 \vee \neg x_8) \wedge$
- $(x_1 \vee x_8 \vee x_{12}) \wedge$
- $(x_2 \vee x_{11}) \wedge$
- $(\neg x_7 \vee \neg x_3 \vee x_9) \wedge$
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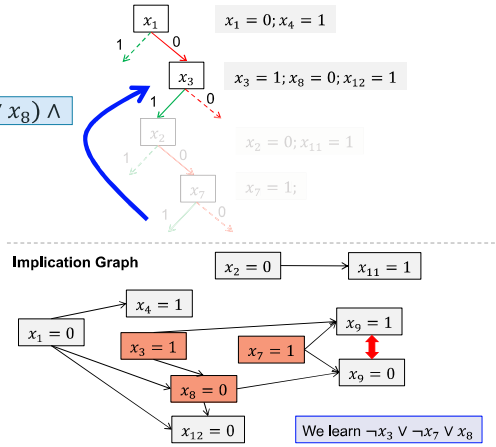

First-Order Logic

PL	FOL
PL assumes the world consists of **facts** that do or do not hold, i.e., true or false	FOL assumes the world consists of **objects** and **relationships**

- A first-order language is defined by
  - (1) Symbols for objects, e.g., $a, b, c, ...$
  - (2) N-ary Predicates (relations returning true/false), e.g., $p(t_1, t_2, ..., t_n)$
  - (3) N-ary Functions (relations), e.g., $f(t_1, t_2, ..., t_n)$

  Terms (t)

  - (4) Formula (F): $\top \mid \bot \mid t \mid \neg F \mid F_1 \vee F_2 \mid F_1 \wedge F_2 \mid F_1 \Rightarrow F_2 \mid F_1 \Leftrightarrow F_2 \mid \forall x.F \mid \exists x.F$
- A first-order language is often used with a "theory" that defines symbols and relationships

Theory of Bit Vectors

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- A first-order language is defined by
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- A first-order language is often used with a "theory" that defines symbols and relationships

Each symbol is a bit vector of a fixed width.

Theory of Bit Vectors

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- A first-order language is often used with a "theory" that defines symbols and relationships

Relational/Logic operations between bitvectors, e.g., *slt*, *ult*, $\wedge$, $\vee$ …

Theory of Bit Vectors

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PL assumes the world consists of **facts** that do or do not hold, i.e., true or false	FOL assumes the world consists of **objects** and **relationships**

- A first-order language is defined by
  - (1) Symbols for objects, e.g., $a, b, c, \ldots$
  - (2) N-ary Predicates (relations returning true/false), e.g., $\ldots$
  - (3) N-ary Functions (relations), e.g., $f(t_1, t_2, \ldots, t_n)$
  - (4) Formula (F): $\top \mid \bot \mid t \mid \neg F \mid F_1 \vee F_2 \mid F_1 \wedge F_2 \mid F_1 \Rightarrow F_2 \mid F_1 \Leftrightarrow F_2 \mid \forall x, F \mid \exists x, F$
- A first-order language is often used with a "theory" that defines symbols and relationships

Arithmetic operations, e.g., +, -, *, /, %, etc.

Bit vector's "+" is different. :>

Theory of Bit Vectors

- How can we check the satisfiability of a constraint?

```
1.  void foobar(int a, int b) {
2.     int x = 1, y = 0;
3.     if (a != 0) {
4.        y = 3+x;
5.        if (b == 0)
6.           x = 2*(a+b);
7.     }
8.     assert(x-y != 0);
9.  }
```

$\alpha_a \neq 0 \wedge \alpha_b = 0 \wedge \neg(2(\alpha_a + \alpha_b) - 4 \neq 0)$

32-bit bitvector for the type int

Satisfiability Modulo Theories

- For the theory of bit vectors
  - **Step 1**: Word-Level Preprocessing
  - **Step 2**: Bit-Blasting (From FOL to PL)
  - **Step 3**: DPLL/CDCL

Example of Step 1:

$\alpha_a \neq 0 \wedge \alpha_b = 0 \wedge \neg(2(\alpha_a + \alpha_b) - 4 \neq 0)$ —Step 1→ $\alpha_a \neq 0 \wedge \alpha_b = 0 \wedge \alpha_a = 2$

—Step 1→ $\alpha_b = 0 \wedge \alpha_a = 2$

Satisfiability Modulo Theories

- For the theory of bit vectors
  - **Step 1**: Word-Level Preprocessing
  - **Step 2**: Bit-Blasting (From FOL to PL)
  - **Step 3**: DPLL/CDCL

Example of Step 2:

$x = [a_7, a_6, \ldots, a_0], y = [b_7, b_6, \ldots, b_0]$ and $z = [c_7, c_6, \ldots, c_0]$ are three 8bit bit vectors

$x \mid y = z \xrightarrow{\text{Step 2}} \bigwedge_{i=0}^{7}((a_i \vee b_i) \Leftrightarrow c_i)$

Satisfiability Modulo Theories

- For the theory of bit vectors
  - **Step 1**: Word-Level Preprocessing
  - **Step 2**: Bit-Blasting (From FOL to PL)
  - **Step 3**: DPLL/CDCL

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$x >_u 0 \xrightarrow{\text{Step 2}} \bigvee_{0}^{7} a_i$ $\qquad$ $x >_s 0 \xrightarrow{\text{Step 2}} \neg a_7 \wedge \bigvee_{0}^{6} a_i$

Summary